

Project Proposal: Reproducing and Extending Zero-Shot Relation Extraction with REPaL

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Abstract

The main problem with relation extraction is its need for large quantities of labelled data, and its difficulty to deal with relations it has not encountered before. REPaL (Zhou et al., 2024) tackles this issue in the *definition-only zero-shot* setting, where only a natural-language definition of a target relation is provided, through a process of generating training examples for the relation, and sequentially augmenting them with feedback. This proposal explains the plan to replicate the results reported in REPaL on their two original test sets (FewRel and WikizSL) and to examine the generalizability of the reported results beyond these sets. The proposed extension generalizes REPaL to five new relation extraction datasets not tested on yet, and to a new, scientifically built relation extraction resource, created from abstracts from the arXiv. There is also a proposal made for an initial plan for an evaluation using LLM-as-a-judge for the complementary evaluation. All the results of the experimental work reported here are tentative and future work not actual results; Section 2.4 describes the current partial results from a first experiment to run the released code.

1 Description of Paper

1.1 What is Relation Extraction?

Relation Extraction (RE) is one of the basic tasks in Natural Language Processing (NLP) which is designed to find out the semantic relation between two different entities that are mentioned in the text. An *entity* is a real-world object or concept that is mentioned in a sentence (for instance, a person, a company, or a place), and a relation type is a type of semantic relationship between two entities (for instance, *founder_of* or *co-founder_of*). For example, if the sentence is "Steve Jobs co-founded Apple Inc. in 1976", an entity extraction system might be able to extract the two entities, "Steve Jobs" and "Apple Inc.", and a relation

extraction system might be able to extract the relation type "co-founder_of". RE has wide applications, such as building knowledge graphs, improving the performance of search engines, and question-answering systems.

In traditional RE systems, significant amounts of labelled training data are needed, however, it is expensive and time-consuming to gather. However, due to the fact that both models are trained on a fixed set of relation types, the model can generally only recognise a relation type labeled in the training set, so *zero-shot* RE tackles this problem by requesting a model to recognise a relation type which has no labeled training examples for it at test time. REPaL studies a stronger version of this, called *definition-only zero-shot* RE: which is given just a natural-language definition (e.g. "a person who is employed by an organisation"), but no labelled instances of the relation, visible or not.

1.2 The Paper's Contribution

Zhou et al. (2024) propose a novel framework called REPaL (Relation Extraction via Pattern Learning), published at the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP).

REPaL leverages the generative capabilities of Large Language Models (LLMs) to synthesise training data from relation definitions. The framework consists of three components, following the paper's own terminology:

- **Definition-Based Seed Construction:** Given only the relation definition, an LLM generates initial positive examples (sentences containing the target relation). Negative examples are obtained by randomly sampling from an unlabelled corpus, under the assumption that the target relation is rare within it.
- **Pattern Learning with a Small Language Model (SLM):** A smaller, bidirectional lan-

guage model, adapted from a natural language inference (NLI) model, is fine-tuned on the synthesised positive and negative seeds to act as a binary classifier for the target relation.

- **Feedback-Driven Instance Improvement and Bias Rectification:** The fine-tuned SLM is run over the unlabelled corpus, and a sample of its predictions is fed back to the LLM. Using its multi-turn conversation ability, the LLM inspects this feedback together with its own synthesis history to generate additional positive examples covering new patterns, and to derive new negative-relation definitions targeting the SLM’s specific errors. The SLM is then retrained on the expanded instance set, and this cycle can repeat for further iterations.

In summary, the pipeline moves from (1) a single relation definition, to (2) LLM-synthesised candidate examples, to (3) a trained SLM classifier, to (4) feedback-driven refinement of that classifier, to (5) the final relation extractor used for inference on new text. Crucially, no stage requires human-labelled data – only the relation definition and an unlabelled corpus. The paper reports that this approach substantially improves zero-shot performance compared to the zero-shot baselines it evaluates against.

1.3 Why This Paper is Suitable for a Novel Project

- **Public code and data:** The authors have released the complete codebase on GitHub, along with the data needed to reconstruct their evaluation sets.
- **Clear experimental setup:** The paper presents well-defined experiments with quantitative results (Precision, Recall, F1) that can be reproduced or compared against.
- **Extension opportunities:** Because the method only requires a relation definition and an unlabelled corpus, it can in principle be applied to new relation types and new domains without retraining from scratch, enabling investigation of generalisation.
- **Established venue:** EMNLP is a major annual conference in empirical NLP research.

2 Explanation of Evaluation

2.1 The Datasets Used in the Original Paper

REPaL is evaluated on two datasets, both derived by the authors from existing relation extraction resources: **DefOn-FewRel** (built from FewRel) and **DefOn-WikiZSL** (built from WikiZSL).

2.1.1 FewRel

FewRel (Han et al., 2018) is a large scale relation classification dataset, which was originally designed for few-shot learning, using sentences from Wikipedia. The public release of its version is 80 relations, each having 700 instances annotated. To create their zero-shot evaluation set, Zhou et al. (2024) randomly sample five groups of 14 relations, respectively, from this public portion as held-out test relations, and then use the rest of the relations’ definition and instances as the target to be evaluated. This derived test collection is referred to as DefOn-FewRel.

2.1.2 WikiZSL

WikiZSL (Chen and Li, 2021) is a zero-shot relation extraction dataset built from Wikidata via distant supervision, consisting of 113 relation types and 93,383 instances (filtered to exclude “none” relations and relations occurring fewer than 300 times). Zhou et al. (2024) sample three groups of 15 relations each from WikiZSL as held-out test relations to construct DefOn-WikiZSL, analogous to the FewRel-based construction above.

For both datasets, the unlabelled corpus used during synthesis and feedback is built by down-sampling 10,000 instances (per test group) from a separate large-scale distantly supervised corpus originally compiled for few-shot relation learning.

2.2 Evaluation Metrics and Procedure

REPaL’s evaluation procedure is a per-relation cross validation. When the test set has R relations, the evaluation is performed R times: for each iteration, the evaluation relation is selected as the target (gold positive) relation, and the rest of the test relations are used as negative (under the assumption of the paper that the negative relation space is not known at test time). The Precision, Recall and F1 are then calculated for that iteration, and the reported scores correspond to the average of the R iterations. The authors therefore also train a separate multi-way classifier (KnowPrompt) on the synthesised data of REPaL and report its Macro-F1 score, which is

compared with the scores of the several baselines (see below) that do require a traditional multi-way classification setup.

The paper compares REPAL against: a random-guess baseline; a GPT-3.5 baseline and a QA-reformulated prompting baseline (RE as QA) applied directly as zero-shot classifiers; a baseline with a RoBERTa backbone applied in both the fully-supervised and zero-shot settings as an NLI classifier (the same backbone on which REPAL fine-tunes); and three previous zero-shot RE models trained on seen labelled data (ZS-BERT, Relation-Prompt, RE-Matching).

Illustrative worked example: To illustrate how Precision, Recall, and F1 relate to each other for a single binary classification outcome (independent of REPAL’s specific per-iteration averaging): suppose a model predicts 100 positive instances for one target relation, of which 80 are correct (precision = 0.80), out of 100 true positive instances in that iteration, of which it correctly identifies 80 (recall = 0.80). The F1 score for that iteration would be

$$F1 = 2 \times \frac{0.80 \times 0.80}{0.80 + 0.80} = 0.80. \quad (1)$$

Reproduction plan: Given the API costs involved in LLM-based synthesis (the paper reports approximately \$3.7 in GPT-4o usage to generate one DefOn-FewRel split’s training and validation instances), I plan to first reproduce REPAL’s Precision/Recall/F1 on one or two DefOn-FewRel splits before attempting DefOn-WikiZSL or the new datasets in Section 4, so that cost and compute are validated on the smaller of the two original settings first.

2.3 Initial Thoughts on LLM-as-a-Judge Evaluation

To complement the extension portion of this project, an LLM-as-a-judge evaluation is proposed in order to evaluate the quality of the extracted relations as an additional analysis. The concept is that a given sentence can be used to infer a relation between two entities and this relation can be semantically valid or not; an LLM can be given to make a binary judgment about whether the relation is valid or not given the context of the sentence. This would give an alternative perspective to the automatic Precision/Recall/F1 metrics, which may not account for the semantic appropriateness of the examples, and might be used to evaluate REPAL’s own generated

training examples.

A candidate design, with agreement between the LLM judge and the human-annotated gold labels (on a subset of 50-100 examples) measured by Cohen’s Kappa, would use a capable general purpose LLM (e.g., GPT-4) through API by calling the LLM as the judge. This is early design and not a final design – the judge model and prompt format and sample size are subject to change if API access and budget is agreed upon, and there are no LLM-as-judge results as of the proposal stage.

2.4 Preliminary Execution and Feasibility Check

I forked and cloned the official GitHub repository and have since made substantive progress on setup and a first partial execution. The following was verified directly:

- **Repository:** <https://github.com/ManoharNaidu/REPAL> (forked from the original)
- **Code:** Written for Python 3.8, with a `requirements.txt` file for dependencies.
- **Data:** Available via Google Drive.
- **Execution:** The repository originally provided two shell scripts for the two training stages. These scripts turned out to contain unfilled placeholder values and assumed a four-GPU setup with correspondingly large batch sizes, so they have since been replaced with a single Python driver script that runs both stages, iterates automatically over all evaluation splits, and collects and aggregates the results.

Driver script: The replacement driver reproduces the same two training stages and hyperparameters the original scripts specified, but adds automatic batch-size adjustment depending on whether a GPU is available (see below), per-run logging, and automatic collation of results across splits into summary and aggregate result files – functionality the original scripts did not provide.

Environment: A Python 3.11.9 virtual environment (`.venv311/`) has been set up with all dependencies from `requirements.txt` installed successfully, including `numpy==1.23.5`, `torch==2.4.0`, `transformers==4.43.3`,

`openai==1.37.1`,
`sentence_transformers==3.0.1`,
and `spacy==3.7.5`. Two additions to `requirements.txt` were needed beyond the authors’ original file: `tensorboard`, which is required by the training scripts but not listed, and a version pin on `httpx` to a release older than 0.28, since the installed `openai` version is incompatible with the newest `httpx` releases and every run would otherwise fail immediately at start-up. This fixes a previous version of this section that stated there was no prebuilt wheel for `numpy==1.23.5` for Python versions newer than 3.10: that is not true and `numpy==1.23.5` installs cleanly with Python 3.11. The problem only affects Python 3.12 and higher due to the requirement for `pkgutil`. This is an API that was removed from Python 3.11+ versions without a prebuilt wheel to fall back on: `ImpImporter`. It isn’t a problem at all when using a Python 3.11 environment, as is done here, and doesn’t require any non-corrective deviation from the authors’ intended target Python 3.8.

Data and model assets: All eight DefOn evaluation splits described in Section 2.1 are downloaded and in place under `data/`: `fewrel_defon_1` through `fewrel_defon_5`, and `wikizsl_defon_1` through `wikizsl_defon_3`, each containing `train.json`, `val.json`, `test.json`, `distant.json`, `rel2id.json`, `id2rel.json`, and `rel_info_updated.json`. Their pre-computed GPT-4o synthesis history and cached seed tensors for all 8 splits are also downloaded to the authors’ `reproduce_main_data/` bundle, and their `roberta-large-mnli` SLM backbone is cached locally.

3 Justification for Paper Choice (Novel Project)

3.1 Research Quality and Reproducibility

Researchers from the University of Illinois Urbana-Champaign and the University of Virginia published REPAL at the empirically oriented NLP research field’s largest annual conference, EMNLP 2024. In the paper the task is precisely and well stated (no more than a definition, zero-shot RE), it is benchmarked against seven baseline methods across random guessers, prompting based methods and prior zero-shot RE and it reports on an ablation

study to isolate the contribution of each of the components of the framework (Section 2). All code is made public on GitHub alongside a description of how to get the evaluation data and the methodology is documented in enough detail, SLM backbone, training hyperparameters and LLM synthesis prompts, to enable a true "reproduction attempt."

3.2 Practical Feasibility and Extension Potential

Fine-tuning the SLM backbone is a small and reasonable-size task that can be done on a single GPU, but will not be done on the local machine due to the time required to complete a full evaluation split as reported in Section 2.4 as would be unfeasible for a CPU in reasonable time. The primary practical expense of the extension is the usage of the LLM API for instance synthesis: the paper estimates this cost to be about \$3.7 per DefOn-FewRel split, and so the costs of the extension are likely to be fairly low if there are only a few splits, which has yet to be independently confirmed. This cost is not charged when DefOn-FewRel and DefOn-WikiZSL are reproduced via the authors’ cached synthesis bundle (Section 2.4) as it does not involve any new API calls; it is deferred to the Section 4 extension datasets, which depend on the newly synthesized amount of credit for relation types not covered by the authors’ bundle and will require fresh synthesis, thus necessitating reclaiming of the OpenAI account used. The environment and dependency issues reported in section 2.4 have been resolved, data access and the SLM backbone have been set up, and a full end-to-end run over a whole evaluation split (with the results aggregated in Precision/Recall/F1) is the final outstanding step. By only needing a relation definition and an unlabelled corpus to be pointed at, REPAL is structurally well suited to being pointed at new relation schemas and domains without the need to retrain a multi-way classifier from scratch, which is used in the generalisation tests in Section 4.

4 New Data

4.1 Existing Datasets Not Used in the Original Paper

The following two datasets, which are not used by Zhou et al. (2024) can be used to test REPAL’s generalisation. They were selected because of their small size and ease of adaptation: both are sentence-level (as opposed to document-level alternatives,

which would require additional work to be done in order to satisfy REPaL’s sentence-level, definition-only input format). See Table 1.

The two datasets must be transformed to the REPaL input format, which is a natural language relation definition and positive and negative instances at sentence level. The compatibility of the input format of the two datasets with the binary NLI-style input format of REPaL has not been verified for either, and will be addressed during pre-processing; existing gold annotations can be used as the ground truth for evaluating the two: the relation schema can be mapped onto natural-language definitions in REPaL’s format.

4.2 New Dataset to Be Created

I propose to construct a small “Scientific Literature Relation Extraction” resource with the following components:

- **Corpus:** An unlabelled collection of 300–500 abstracts collected from arXiv papers in computer science and related fields. This corpus plays the same role as the unlabelled corpus REPaL uses during synthesis and feedback – it is not annotated.
- **Relation schema:** 5–10 scientific relations (e.g., `proposes_method`, `extends_prior_work`, `evaluates_on_dataset`, `achieves_performance`, `compares_with_baseline`), each expressed as a natural-language definition in the same style REPaL expects.
- **REPaL input:** Only the relation definitions and the unlabelled abstract corpus are given to REPaL, consistent with the definition-only zero-shot setting – no human-labelled examples are provided to the framework at any stage.
- **Human annotation (evaluation only):** A separate subset of 50–100 abstracts, distinct from what REPaL sees, is proposed to be independently annotated by 2–3 annotators to build gold-standard evaluation labels. Disagreements will be resolved through discussion to produce a consensus gold standard, and inter-annotator agreement will be measured with Cohen’s Kappa before consensus is applied.

- **Evaluation:** REPaL’s output on the 50–100 annotated abstracts will be compared against this human-annotated gold standard, following the same Precision/Recall/F1 metrics described in Section 2.

The datasets need to be translated from their native format to one that REPaL expects: the relation definition in natural language and positive and negative sentences at sentence level. In order to test for compatibility with REPaL’s binary NLI-style input format, it will be necessary to verify the compatibility for both data sets in the preprocessing phase; existing gold annotations can be used as the evaluation ground truth for the use of the relation schema and mapping it to natural-language definitions written in REPaL’s format.

5 Research Questions and Project Plan

5.1 Research Questions

Primary question: Does the REPaL framework for definition-only zero-shot relation extraction generalise to new relation types and domains beyond those evaluated in the original paper?

Core secondary questions (directly required to answer the primary question):

- Does REPaL maintain its effectiveness when applied to the new datasets in Section 4.1 and the scientific-literature domain in Section 4.2, compared to its reported DefOn-FewRel/DefOn-WikiZSL performance?
- What types of relation definitions lead to the best synthesis quality, and does this differ across domains?

Exploratory questions (time and resources permitting):

- How many synthesis iterations are needed before performance saturates on a new domain, compared to the original paper’s own iteration analysis? The released code fixes this iteration count at one, so answering this question required a small code change to expose it as an adjustable setting.
- Does an LLM-as-a-judge (Section 2.3) agree with the automatic Precision/Recall/F1 metrics?

Given the ten-week timeline: **must complete** is reproduction on both DefOn-FewRel and DefOn-WikiZSL, evaluation on both new datasets, and

Dataset	Source	Relation Types	Size	Why Relevant
SemEval-2010 Task 8	HuggingFace	19 directional relations	10,000+ sentences	Standard RE benchmark with a different relation schema
BioCreative CDR	V NCBI	Chemical-disease relations	1,500 articles	Biomedical domain; tests cross-domain generalisation

Table 1: Existing datasets proposed for testing REPaL’s generalisation beyond the original paper.

Week	Activity	Deliverable
1	Submit Project Proposal; migrate to GPU compute and run the full reproduction driver across all 5 DefOn-FewRel splits	This document; DefOn-FewRel results vs. paper
2	Run the full reproduction driver across all 3 DefOn-WikiZSL splits and compare both datasets’ results against the paper’s reported Precision/Recall/F1	DefOn-WikiZSL results vs. paper
3	Download and preprocess new datasets (SemEval-2010 Task 8, BioCreative V CDR)	Processed data
4	Run REPaL on new datasets	New dataset results
5	Construct scientific literature dataset (corpus, relation schema, human annotation)	Annotated corpus
6	Run REPaL on scientific literature dataset	New data results
7	Conduct LLM-as-judge evaluation	LLM evaluation results
8	Statistical analysis and error analysis	Analysis notebooks
9	Update presentation; begin final report writing	Presentation slides; report draft
10	Final report due	Final submission

Table 2: Project plan and timeline (ten weeks).

the scientific-literature dataset; **should complete** is the LLM-as-a-judge analysis and a detailed error analysis; **stretch goal** is the iteration-saturation analysis.

5.2 Project Plan and Timeline

The project plan is presented in Table 2 week-by-week, in a shorter version that used to be fifteen weeks. This compression starts from the point when the reproduction runs are completed instead of when they are set up because environment setup, data download, and API configuration are assumed to be finished as reported in Section 2.4. The next challenging week is Week 5 (constructing the scientific-literature dataset), where all the previous days are now condensed into a single week: corpus collection, relation-schema definition, and the human-annotation subset.

6 Conclusion

This proposal has proposed a framework to replicate and extend the REPaL framework for definition-only zero-shot relation extraction. The code and data are available, and Section 2.4 reports on real forward-looking progress using the code released: reported problems with dependencies are now fixed, and the pipe is working as described by the authors, with their cache of synthesis data. The best thing is to perform a complete evaluation run with appropriate GPU hardware and report the metrics. The project aims at replicating the original results from REPaL, using it on two existing datasets it was not evaluated on, building anew a scientific-literature dataset, and examining if its claimed performance extends beyond the original context. This is consistent with the objectives of reproducible research, computational experimentation and open science that are part of the goals of COMP8240, and full feasibility will be confirmed when the outstanding execution step above is completed.

References

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